The documentation and process conversion measures necessary to comply with this revision shall be completed by 9 January 2007.

INCH POUND

MIL-PRF-19500/472E
9 October 2007
SUPERSEDING
MIL-PRF-19500/472D
9 May 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, POWER, TYPES 2N6350, 2N6351, 2N6352, AND 2N6353, JAN, JANTX, JANTXV, JANHC, AND JANKC

Inactive for new design after 13 March 2001.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN Darlington, silicon power transistors. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-33) for 2N6350 and 2N6351, figure 2 (3-pin TO-66) for 2N6352 and 2N6353, and figure 3 for JANHC and JANKC (die).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T T_A = +25°C	P_T T_C = +100°C	$R_{\theta JC}$ Max	V_{EB1}	V_{EB2}	V_{CBO1}	V_{CER}	I_C	I_C (1)	I_{B1}	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	V dc	V dc	A dc	A dc	A dc	°C
2N6350	1.0 (2)	5 (3)	20	12	6	80	80	5	10	0.5	-65 to +200
2N6351	1.0 (2)	5 (3)	20	12	6	150	150	5	10	0.5	-65 to +200
2N6352	2.0 (4)	25 (5)	4.0	12	6	80	80	5	10	0.5	-65 to +200
2N6353	2.0 (4)	25 (5)	4.0	12	6	150	150	5	10	0.5	-65 to +200

(1) Applies for $t_p \leq 10$ ms, duty cycle ≤ 50 percent.
(2) Derate linearly 5.72 mW/°C for T_A > +25°C.
(3) Derate linearly 50 mW/°C for T_C > +100°C.
(4) Derate linearly 11.4 mW/°C for T_A > +25°C.
(5) Derate linearly 250 mW/°C for T_C > +100°C.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise specified.

Limit	h_{FE} (1) V_{CE} = 5 V dc I_C = 5 A dc		$\|h_{fe}\|$ (1) V_{CE} = 10 V dc I_C = 1 A dc f = 10 MHz	$V_{BE(ON)1}$ V_{CE} = 5 V dc, I_C = 5 A dc	C_{OBO} V_{CB1} = 10 V dc, 100 kHz ≤ f ≤ 1 MHz,	$V_{CE(SAT)}$		Switching V_{CC} = 30 V dc I_C = 5 A dc	
						R_{B2E} = 100 Ω I_C = 5 A dc I_{B1} = 5 mA dc	R_{B2E} = 100 Ω I_C = 5 A dc I_{B1} = 10 mA dc		
	2N6350 2N6352	2N6351 2N6353			base 2 open	2N6350 2N6352	2N6351 2N6353	t_{on}	t_{off}
				V dc	pF	V dc	V dc	µs	µs
Min	2,000	1,000	5						
Max	10,000	10,000	25	2.5	120	1.5	2.5	0.5	1.2

(1) Pulsed, see 4.5.1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

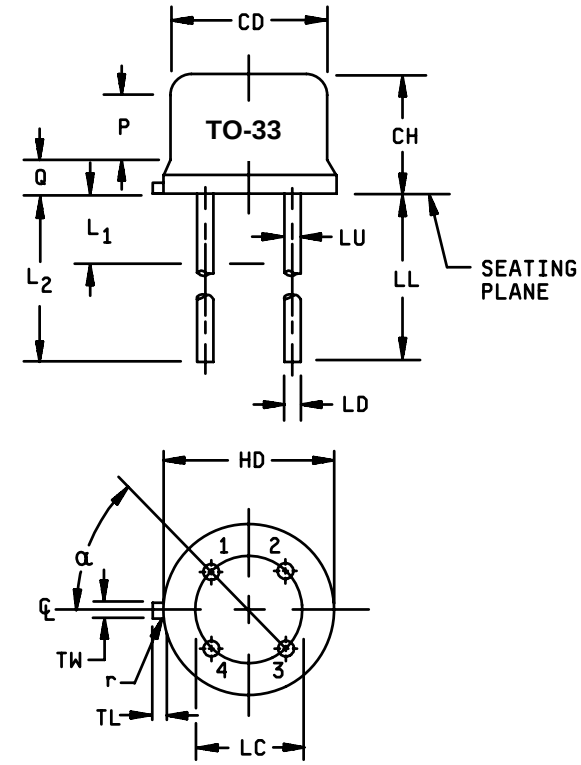
2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

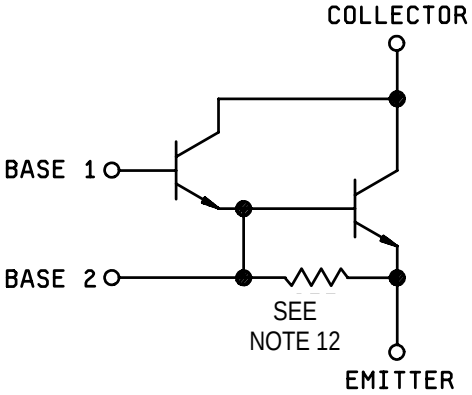
3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

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SCHEMATIC CIRCUIT

* FIGURE 1. Dimensions and configuration for 2N6350 and 2N6351 (TO-33).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	6
CH	.184	.260	4.67	6.60	
HD	.335	.370	8.51	9.40	
LC	0.200 TP		5.080 TP		7
LD	.016	.021	0.407	0.533	8, 9
LL	1.500	1.750	38.10	44.45	8, 9
L_1		.050		1.27	8, 9
L_2	.250		6.35		8, 9
LU	.016	.019	0.407	0.482	8, 9
TL	.029	.045	0.74	1.14	4
TW	.028	.034	0.712	0.863	3
P	.100		2.54		6
Q		.050		1.27	5
r		.010		0.254	11
α	45° TP		45° TP		7

NOTES:

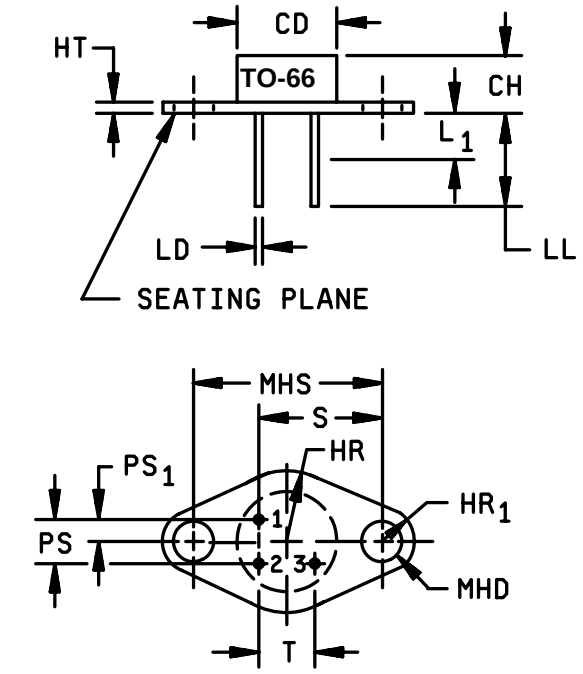
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 –.000 inch (1.37 +0.03 –0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
8. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All leads.
10. Lead designation is as follows: 1 - Emitter, 2 - Base (B2), 3- Base (B1), 4 - Collector. The collector shall be connected to the case.
11. r (radius) applies to both inside corners of the tab.

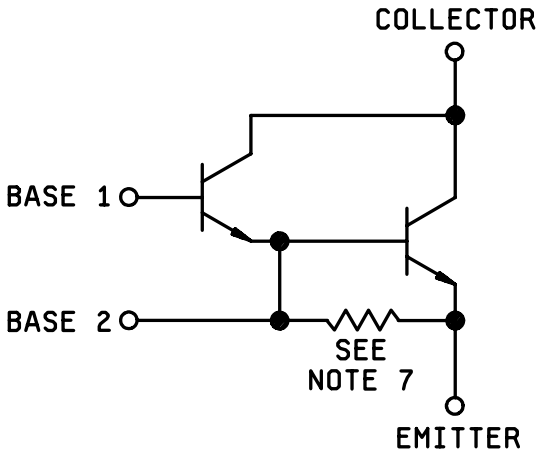
* 12. Internal resistance (typically 750 ohms). This resistor is optional.

13. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Dimensions and configuration for 2N6350 and 2N6351 (TO-33) - Continued.

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SCHEMATIC CIRCUIT

* FIGURE 2. Physical dimensions for 2N6352 and 2N6353 (3-pin TO-66).

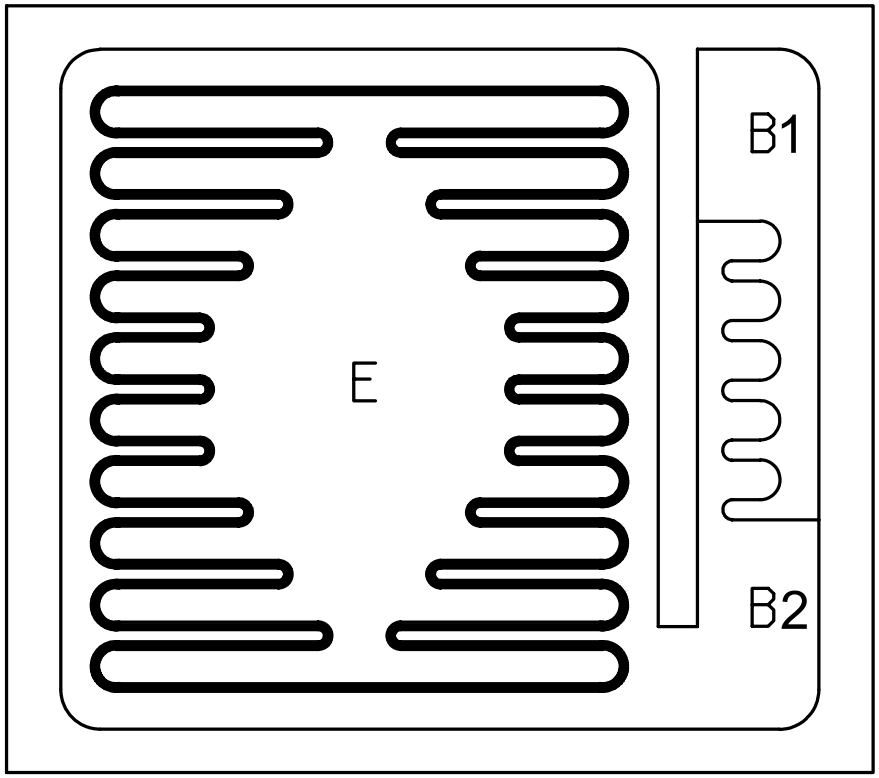
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.620		15.75	
CH	.250	.340	6.35	8.64	
HT	.050	.075	1.27	1.91	4
HR		.350		8.89	
HR_1	.115	.145	2.92	3.68	
LD	.028	.034	0.711	0.863	5
LL	.360	.500	9.14	12.70	5
L_1		.050		1.27	5
MHD	.142	.152	3.61	3.86	
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	
PS_1	.093	.105	2.36	2.67	
S	.570	.590	14.48	14.99	
T	.190	.210	4.83	5.33	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The outline contour is optional.
4. Dimension does not include sealing flanges.
5. All leads.
6. Lead designation is as follows: 1 - Emitter, 2 - Base (B1), 3- Base (B2), Collector. The collector shall be connected to the case.
* 7. Internal resistance (typically 750 ohms). This resistor is optional.
8. Shape of capweld flange is optional and cannot extend beyond HR.
9. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 2. Physical dimensions for 2N6352 and 2N6353 (TO-66) - Continued.

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NOTES:

1. Chip size: 87 x 100 mils ±2 mils.
2. Chip thickness: 10 ±1.5 mils nominal.
3. Top metal: Aluminum 30,000Å minimum, 33,000Å nominal.
4. Back metal: A. Al/Ti/Ni/Ag 12kÅ/3kÅ/7kÅ/7kÅ min.,15kÅ/5kÅ/10kÅ/10kÅ nom.
   B. Gold: 2,500Å minimum, 3,000Å nominal.
5. Backside: Collector.
6. Bonding pad: B_1 = 16 x 14 mils, B_2 = 16 x 18 mils, E = 40 x 8 mils.

* FIGURE 3. JANHC and JANKC A-version die dimensions.

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-33) for 2N6350 and 2N6351, figure 2 (3-pin TO-66) for 2N6352 and 2N6353, and figure 3 for JANHC and JANKC (die) herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2)

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels only
(1) 3c	Thermal impedance (see 4.3.2).
11	I_{CEX1} and h_{FE1}
12	See 4.3.1
13	ΔI_{CEX1} = ±100 percent of initial value or 20 nA dc, whichever is greater; Δh_{FE1} = ±25 percent of initial value; subgroup 2 of table I herein.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = +25°C; Power shall be applied to achieve T_J = +135°C minimum and power dissipation of $P_D \geq 75$ percent of maximum rated P_T as defined in 1.3.

2N6350, 2N6352 V_{CE} = 60 V dc
2N6351, 2N6353 V_{CE} = 100 V dc

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. The thermal impedance limit used in screen 3c and table I, subgroup 2 shall be set statistically by the supplier. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be as specified in table III herein.

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4.4.2.1 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	For 2N6350 and 2N6352; V_{CE} = +40 V dc. For 2N6351 and 2N6353; V_{CE} = +100 V dc. Power shall be applied to achieve T_J = +150°C minimum and power dissipation of $P_D \geq 75$ percent of maximum rated P_T as defined in 1.3.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E for 2N6350 and 2N6351. Test condition A for 2N6352 and 2N6353, weight = 3 lbs, t = 15s.
C5	3131	Base 1 shorted to base 2 (see 1.3).
C6	1026	For 2N6350 and 2N6352; V_{CE} = +40 V dc. For 2N6351 and 2N6353; V_{CE} = +100 V dc. Power shall be applied to achieve T_J = +150°C minimum and power dissipation of $P_D \geq 75$ percent of maximum rated P_T as defined in 1.3.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Collector to emitter breakdown voltage	3011	Bias condition B, I_C = 25 mA dc, pulsed (see 4.5.1), R_{B1E} = 2.2 k ohm, R_{B2E} = 100 ohms	$V_{(BR)CER}$			
2N6350, 2N6352				80		V dc
2N6351, 2N6353				150		V dc
Emitter to base 1, breakdown voltage	3026	Bias condition D, I_E = 12 mA dc, base 2 open.	$V_{(BR)EBO1}$	12		V dc
Emitter to base 2, breakdown voltage	3026	Bias condition D, I_E = 12 mA dc, base 1 open.	$V_{(BR)EBO2}$	6		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{EB1} = 2 V dc, R_{B2E} = 100 ohms	I_{CEX1}			
2N6350, 2N6352		V_{CE} = 80 V dc			1.0	µA dc
2N6351, 2N6353		V_{CE} = 150 V dc			1.0	µA dc
Saturated voltage and resistance	3071	I_C = 5.0 A dc; R_{B2E} = 100 ohms	$V_{CE(sat)1}$			
2N6350, 2N6352		I_B = 5 mA dc, pulsed (see 4.5.1)			1.5	V dc
2N6351, 2N6353		I_B = 10 mA dc, pulsed (see 4.5.1)			2.5	V dc
Base emitter voltage (unsaturated)	3066	V_{CE} = 5 V dc, I_C = 5.0 A dc pulsed (see 4.5.1) R_{B2E} = 100 ohms	$V_{BE(on)1}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1.0 A dc; pulsed (see 4.5.1). R_{B2E} = 1 k ohms	h_{FE1}			
2N6350, 2N6352				2,000		
2N6351, 2N6353				1,000		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Forward-current transfer ratio	3076	V_{CE} = 5 V dc I_C = 5.0 A dc; pulsed (see 4.5.1). R_{B2E} = 100 ohms	h_{FE2}			
2N6350, 2N6352 2N6351, 2N6353				2,000 1,000	10,000 10,000	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc I_C = 10 A dc; pulsed (see 4.5.1). R_{B2E} = 100 ohms	h_{FE3}			
2N6350, 2N6352 2N6351, 2N6353				400 200		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A V_{EB1} = 2 V dc; R_{B2E} = 100 ohms V_{CE} = 80 V dc	I_{CEX2}			
2N6350, 2N6352 2N6351, 2N6353		V_{CE} = 150 V dc			1.0 1.0	mA dc mA dc
Low temperature operation:		T_A = -65°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc I_C = 5.0 A dc, pulsed (see 4.5.1) R_{B2E} = 100 ohms	h_{FE4}			
2N6350, 2N6352 2N6351, 2N6353				400 200		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 1.0 A dc f = 10 MHz, R_{B2E} = 100 ohms	$\|h_{fe}\|$	5	25	
Open circuit output capacitance	3236	V_{CB1} = 10 V dc; 100 kHz ≤ f ≤ 1 MHz, base 2 open	C_{obo}		120	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Pulse response						
Turn-on time	3251	Test condition A, V_{CC} = 30 V dc I_C = 5.0 A dc	t_{on}			
2N6350, 2N6352 2N6351, 2N6353		See figure 4 See figure 5			0.5 0.5	µs µs
Turn-off time	3251	Test condition A, V_{CC} = 30 V dc I_C = 5.0 A dc	t_{off}			
2N6350, 2N6352 2N6351, 2N6353		See figure 4 See figure 5			1.2 1.2	µs µs
Subgroup 5						
Safe operating area (continuous dc) (for types 2N6350 and 2N6351 only)	3051	T_C = +100°C; t ≥ 1.0 s, 1 cycle $t_r + t_f$ = 10 µs (see figure 6) R_{B2E} = 100 ohms				
Test 1 (2N6350, 2N6351)		V_{CE} = 1.5 V dc; I_C = 3.3 A dc				
Test 2 (2N6350, 2N6351)		V_{CE} = 30 V dc; I_C = 167 mA dc				
Test 3 (2N6350)		V_{CE} = 80 V dc; I_C = 35 mA dc				
Test 4 (2N6351)		V_{CE} = 150 V dc; I_C = 13 mA dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued.						
Safe operating area (continuous dc) (for types 2N6352 and 2N6353 only)	3051	T_C = +100°C; t ≥ 1.0 s, 1 cycle $t_r + t_f$ = 10 μs (see figure 7) R_{B2E} = 100 ohms				
Test 1 (2N6352, 2N6353)		V_{CE} = 5.0 V dc; I_C = 5 A dc				
Test 2 (2N6352, 2N6353)		V_{CE} = 10 V dc; I_C = 2.5 A dc				
Test 3 (2N6352)		V_{CE} = 80 V dc; I_C = 95 mA dc				
Test 4 (2N6353)		V_{CE} = 150 V dc; I_C = 35 mA dc				
Electrical measurements		See table I, subgroup 2.				
Safe operating area (switching)	3053	Load condition B (clamped inductive load) see figure 8, T_A = + 100°C, $t_r + t_f \leq 1.0$ μs, duty cycle ≤ 2 percent; t_p = 8 ms (vary to obtain I_C) R_s = 0.1 Ω; R_{BB1} = 200 Ω, V_{BB1} = 10 V dc, R_{BB2} = 2.2 kΩ, V_{BB2} = -4 V dc, I_C = 5 A dc, V_{CC} = 25 V dc; $R_L \leq 2.5$ Ω, L = 40 mH (Triad c-48U, or equivalent), R_{B2E} = 100 Ω				
2N6350, 2N6352 2N6351, 2N6353		Clamp voltage = 50 +0, -5 V dc Clamp voltage = 90 +0, -5 V dc				
Electrical measurements		See table I, subgroup 2				

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) - for qualification and requalification only.

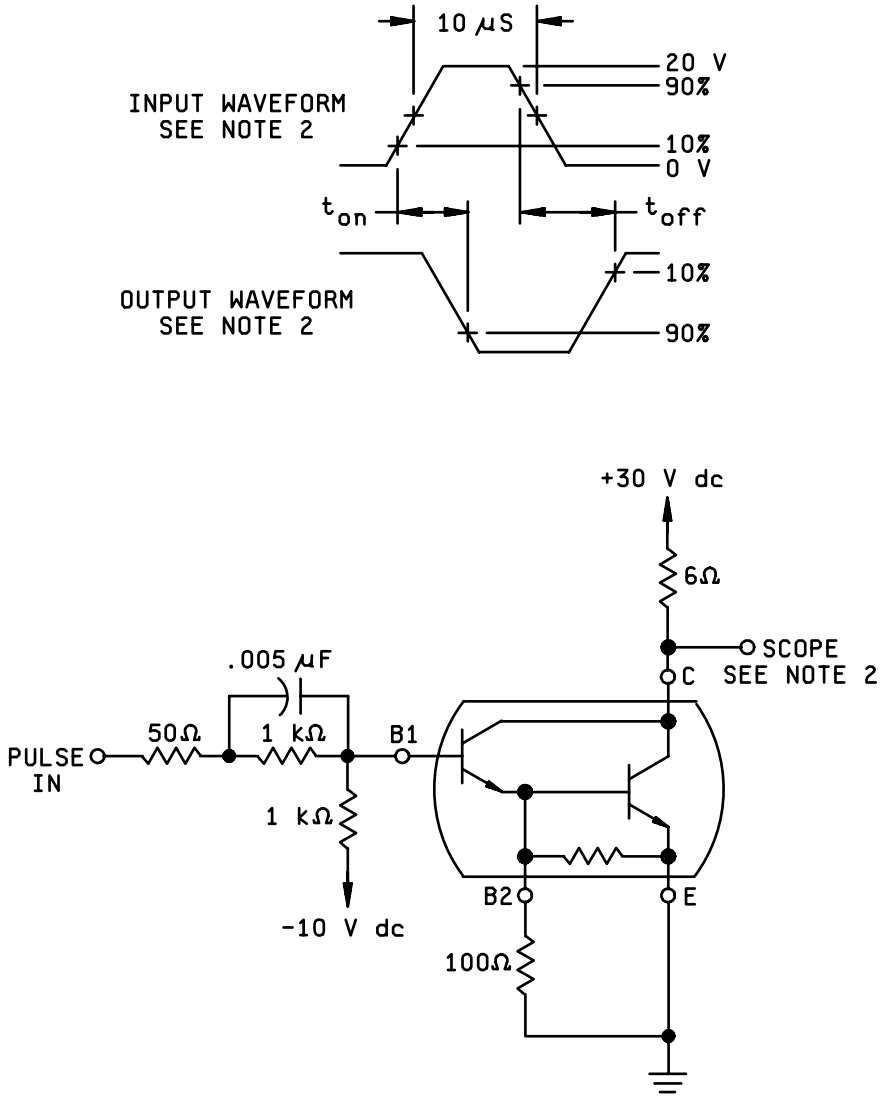
Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
End-point electrical measurements		See table I, subgroup 2 and table III herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles.	
End-point electrical measurements		See table I, subgroup 2 and table III herein.	
Subgroup 4			sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			3 devices
ESD	1020	Testing not required for devices listed as class 3. Testing required for nonsensitive listing to prove capability.	
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

* TABLE III. Delta requirements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limit
		Method	Conditions		
1.	Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 5.0 A dc; pulsed see 4.5.1, R_{B2E} = 100 ohm	Δh_{FE2}	±25 percent change from initial reading.

1/ The delta measurements for table E-VIB (JAN, JANTX and JANTXV) are: Subgroup 3 of MIL-PRF-19500, see table III, step 1 herein.
2/ The delta measurements for table E-IX of MIL-PRF-19500 are: Subgroups 1 and 2, see table III, step 1 herein.

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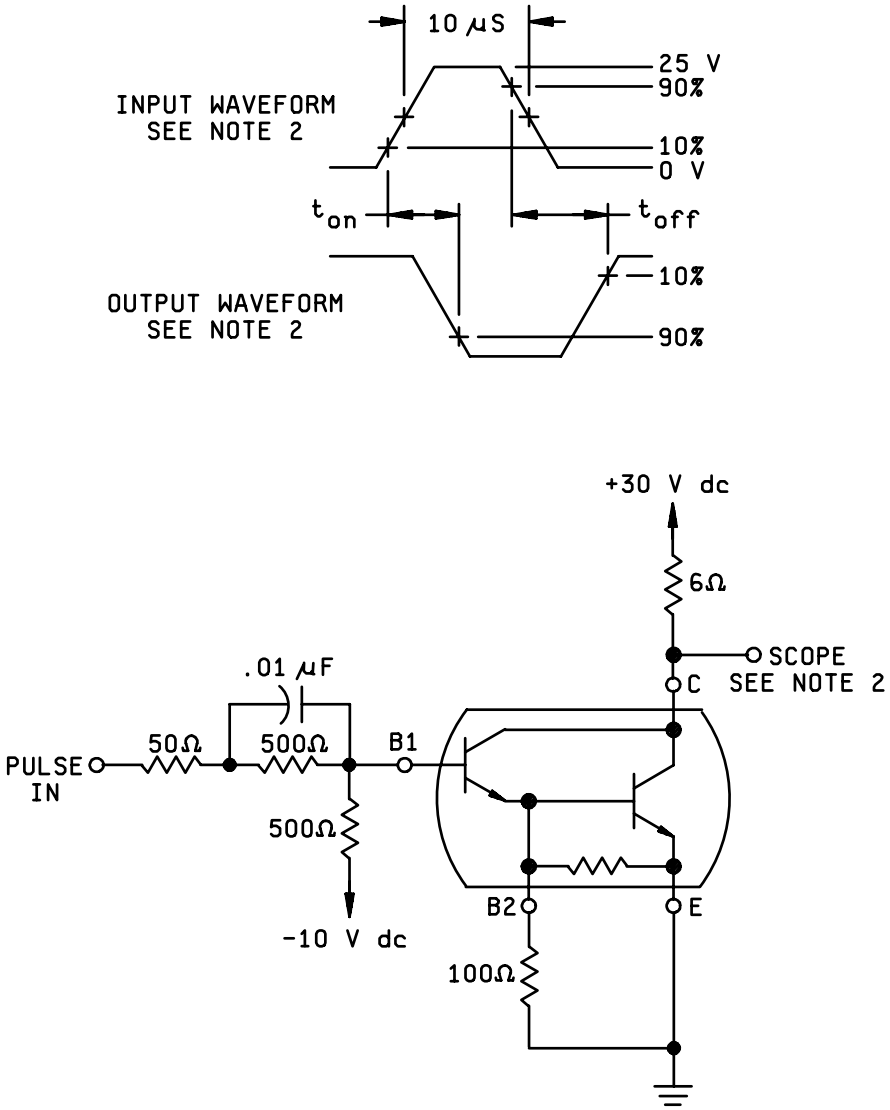


NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{OUT} = 50\ \Omega$, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{in} \geq 10\ M\Omega$, $C_{in} \leq 11.5$ pf.
3. Resistors shall be non-inductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 4. Pulse response test circuit for 2N6350 and 2N6352.

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NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_f \leq 15$ ns, $Z_{Out} = 50\ \Omega$, PW = 10 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $Z_{in} \geq 10\ M\Omega$, $C_{in} \leq 11.5$ pf.
3. Resistors shall be non-inductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 5. Pulse response test circuit for 2N6351 and 2N6353.

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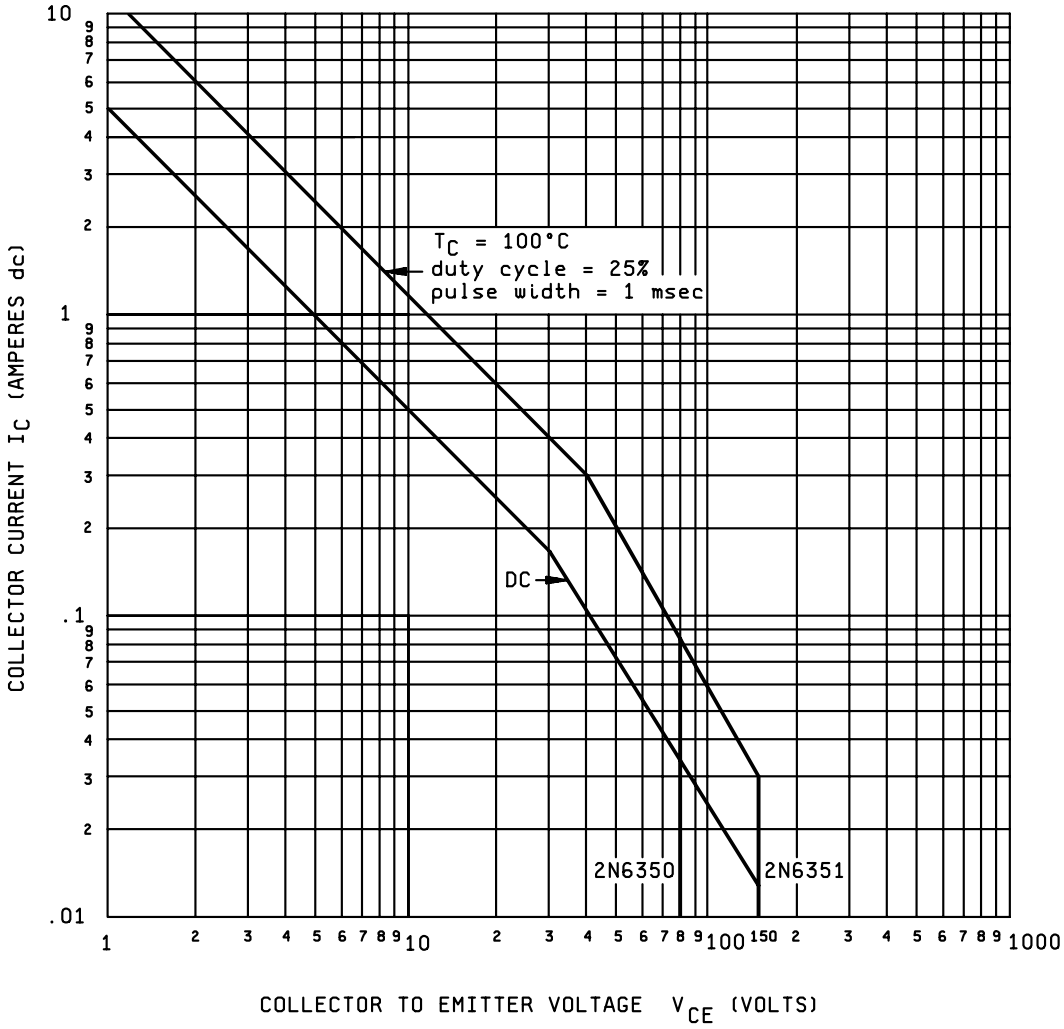


FIGURE 6. Maximum safe operating area graph for types 2N6350 and 2N6351.

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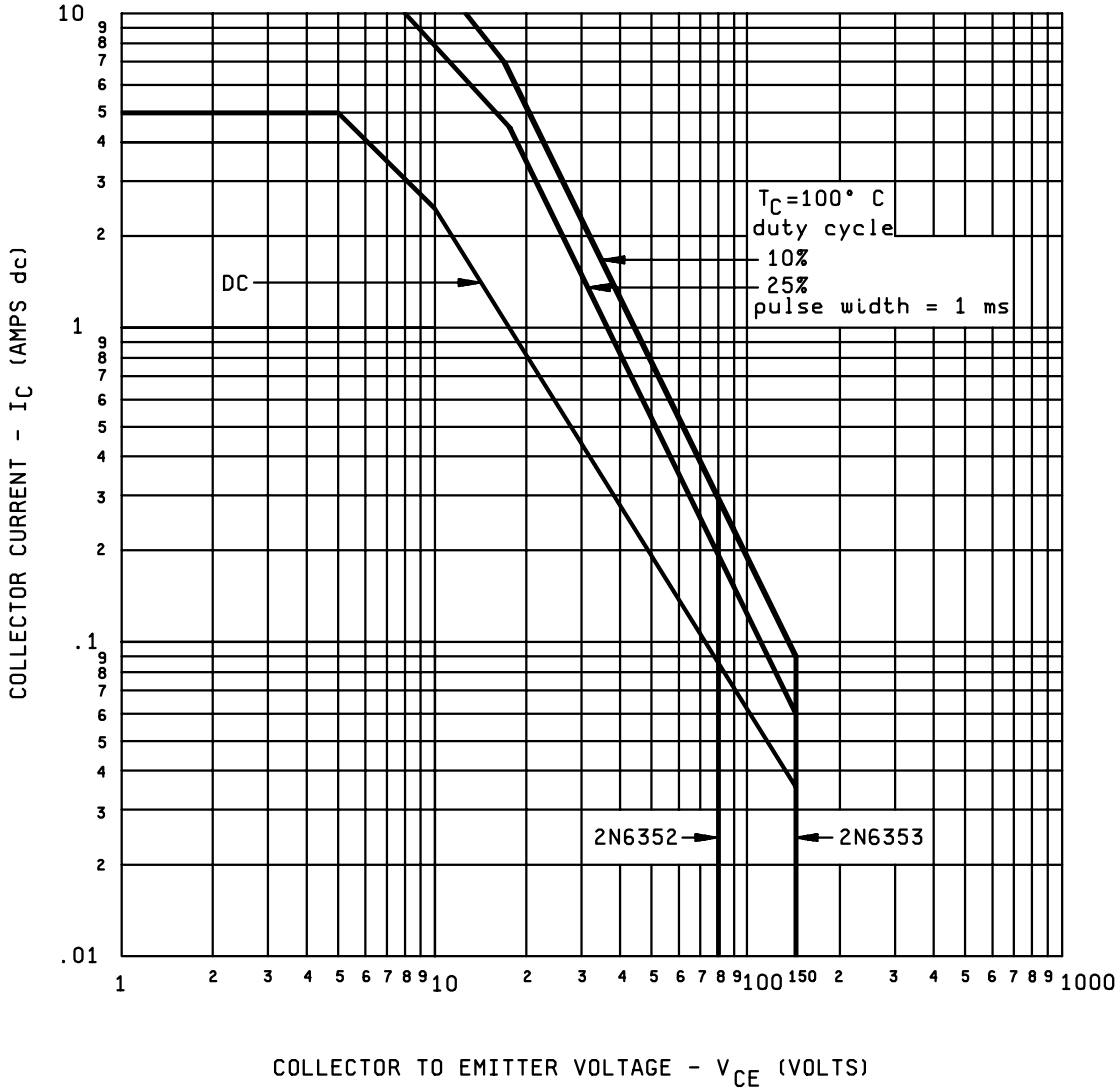


FIGURE 7. Maximum safe operating area graph for types 2N6352 and 2N6353.

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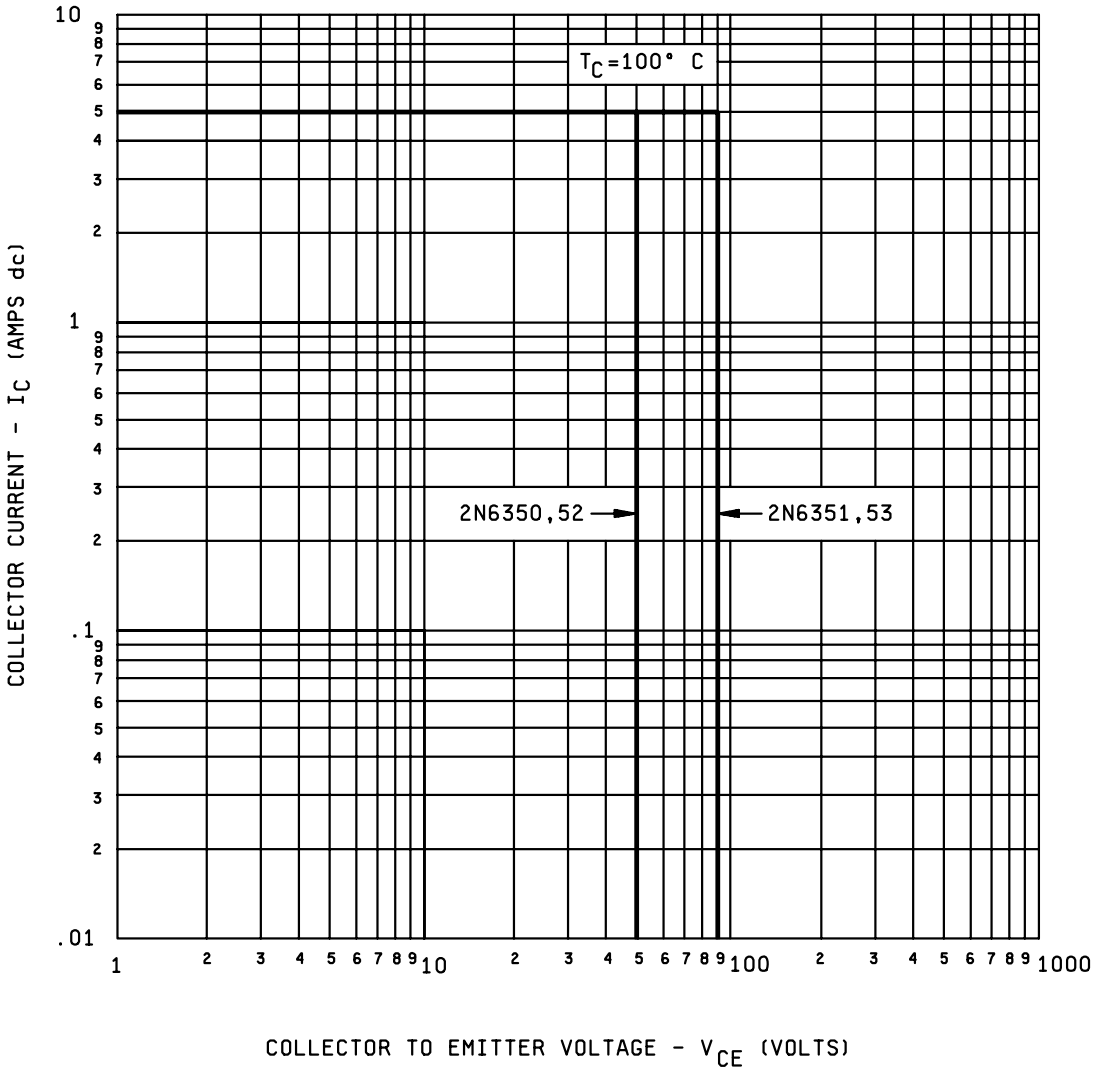


FIGURE 8. Safe operating area for switching between saturation and cutoff – (clamped inductive load).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N6350) will be identified on the QML.

JANC ordering information	
PIN	Manufacturers
	43611
2N6350	JANHCA2N6350 JANKCA2N6350

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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